

# Trax9 AI Lead Generation & Outreach

An automated pipeline that reads the agency's own website to learn what it sells, finds businesses that match that profile, audits each one's site, ads and SEO, then writes and sends outreach that references what it actually found.

Stack FastAPI · PostgreSQL · Celery/Redis · React Pipeline 8 stages Tenancy Multi-user, per-user keys

HOSTED API BROWSER + PROXY AI MODEL DATABASE EMAIL

PIPELINE A lead moves top to bottom. Each stage writes its result back before the next begins.



#### Reads in

- Agency services
- Full audit + top gap
- Social signals

#### Writes out

- Subject naming a real finding
- Three to four paragraphs
- Signed by a named consultant

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### Send Pipeline

Four gates before anything leaves. Any one of them can hold a message back.

#### Rate limits

Hourly and daily, counted in Redis so every worker agrees.

#### Warmup

Starts at 10 a day, climbs by 5 until it reaches the real cap.

#### Suppression

Unsubscribes, spam reports and hard bounces are never contacted again.

#### Quota

Monthly ceiling per account. Overrun pauses the campaign.

Then SendGrid sends it — with an unsubscribe link, a one-click unsubscribe header and a postal address in the footer, as the law requires.

**AFTER THE SEND** Everything that comes back changes what happens next.

#### INBOUND SIGNALS

##### Opens and clicks

Recorded against the lead from SendGrid's event webhook, verified by signature and de-duplicated by event id.

##### Bounces and spam reports

Added to the suppression list automatically. That address is finished.

##### Replies

Detected by polling the inbox, or by inbound parse. Matched back to the original message.

#### FOLLOW-UP SEQUENCE

##### Day 3 — second insight

Not a bump. A different finding from the same audit.

##### Day 7 — close the loop

Short, final, no pressure.

##### Cancels on any reply

A reply, unsubscribe or bounce cancels every remaining step for that lead immediately. Nobody gets chased after answering.

**DASHBOARD** What the operator sees and controls.

#### VIEWS

##### Overview

Leads found, audited, sent, replied — plus daily activity.

##### Campaigns

Create, run each stage, watch live progress.

##### Leads

Sorted by fit score. Filter by status, flag the ones missing an email.

#### LEAD DETAIL

##### Audit tabs

Website · SEO · Meta Ads · Google Ads · Brand · Social

##### Email preview

Editable before it goes. Regenerate on demand.

##### Sequence timeline

Where this lead sits across the first email and both follow-ups.

#### SETTINGS

##### Keys

Each account supplies its own and they are encrypted at rest. Displayed masked, never in full.

##### Sending

Daily and hourly caps, send window, warmup, return address.

##### Spend

Emails, AI usage and paid lookups counted per month against the account's ceiling.

**APPLIES TO EVERY STAGE** These wrap the pipeline rather than sit inside it.

#### ACCESS & SECRETS

##### Accounts

Sign-in issues a token; every request is scoped to the account that made it. No query can reach another account's leads.

##### Encrypted keys

Third-party credentials are encrypted before storage and decrypted only in the moment they are used.

#### STAYING UNBLOCKED

##### Rotation

Every browser run goes out through a rotating proxy and user-agent pool.

##### Backing off

Politeness delays per site; repeated blocks pause that source rather than hammer it.

#### VISIBILITY

##### Progress

Long jobs report item by item, so a running campaign shows real progress and real failures.

##### Health

A health endpoint, structured logs and error reporting across the app and its workers.

#### INFRASTRUCTURE

##### PostgreSQL

SQLAlchemy · Alembic

##### Redis

Rate counters · Queue broker

##### Celery workers

5 queues · profile, discovery, email\_find, audit, email

##### Celery beat

3 jobs · warmup, sequences, replies

##### Docker Compose

Railway · Fly.io · DigitalOcean

Sending runs on an authenticated subdomain with SPF, DKIM and DMARC — separate from the main corporate domain, so outreach volume can never affect normal business mail.